Emergent Heterogeneity in Compartmental Disease Models

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Abstract

Heterogeneity in infectious disease spread can reshape the risk and severity of outbreaks. This heterogeneity can arise from intrinsic individual-level differences in susceptibility, contact patterns and infectious durations. Although intrinsic heterogeneity has been examined in depth, heterogeneity can also arise as a consequence of the disease dynamics. Changes in the size of susceptible populations can lead to emergent heterogeneity, i.e., differences in realized variation in the number of secondary cases caused by a primary case. Here, we investigate the magnitude and mechanisms generating emergent heterogeneity by analyzing secondary case distributions within the canonical SIR model. We first describe κ, a statistic for quantifying the emergent heterogeneity in secondary case counts in the SIR model and stochastic realizations. Although intrinsic heterogeneity decreases systematically with $\mathcal{R}_0$, we show that κ is invariant for all values of $\mathcal{R}_0$ within stochastic SIR models and we prove this result mathematically. The invariance of emergent heterogeneity in SIR models results from distinct partitioning of within- and between-cohort components of κ. Our results demonstrate that heterogeneity emerges even in simple SIR epidemics as a result of feedback between stochastic depletion of susceptibles and subsequent opportunities of transmission. More broadly, our results suggest that efforts to understand the basis for realized variation in transmission should account for the impacts of susceptible depletion beyond the narrow limits of the initial outbreak phase.

Abstract

[**JD:** *I prefer this abstract, but left the other one for discussion.*] Heterogeneity in infectious disease spread can reshape the risk and severity of outbreaks. This heterogeneity can arise from intrinsic individual-level differences in susceptibility, contact patterns and infectious durations. Heterogeneity can also emerge from disease dynamics in a canonically homogeneous population, due to the fact that individuals randomly become infected at different times, and recover after different intervals. Here,

we explore the mechanisms generating emergent heterogeneity by analyzing secondary case distributions within the canonical SIR model. We show that, when we consider "forward cohorts" of individuals infected at the same time, between-cohort variation increases as $\mathcal{R}_0$ increases, while within-cohort variation decreases. We further show that these effects balance exactly, so that individual-based variation in either expected or realized number of secondary cases does not change with $\mathcal{R}_0$. These results draw attention to the importance of considering emergent, as well as intrinsic, variation between individuals, and of considering heterogeneity of spread over the whole epidemic cycle, as well as during the much-better-studied initial outbreak phase.

1 Introduction

[**JD:** *Need to substantially de-emphasize superspreading to follow Josh's vision. Can still leave a little bit in...*] [**JSW:** *Not even sure... I just wanted to get the word superspreading out of the title since we don't quite get there. I really like this motivation!*] Real outbreaks have significant heterogeneity in the number of secondary cases – including superspreading, marked by a small fraction of individuals causing a disproportionately large fraction of realized transmission [13, 19]. Superspreading can help explain why epidemics take off in some situations and not in others, despite apparently similar contexts [12, 4]. Leveraging knowledge of superspreading can make control efforts more effective [5, 15, 18]. It is especially important for disease dynamics and control strategies at the beginning of outbreaks [6]. Correspondingly, theoretical treatment of superspreading emphasizes individual-level transmission patterns, often while assuming a fully susceptible population, [13, 9] and has focused on models that link observed secondary case counts to a continuous random variable, the *individual reproductive number*, via a Poisson sampling process. Canonically, the individual reproductive number is modeled as a gamma distribution, giving rise to negative binomial secondary case distributions [13]. In this way, variation arising from the stochastic transmission process is conceptually separated from *intrinsic heterogeneity* in expected infectiousness that may emerge from potential differences in biology or behaviors between individuals [3, 1, 7].

Conventional models accounting for heterogeneity typically assume a constant pool of susceptible individuals, implicitly or explicitly focusing on the very beginning of an outbreak. Over longer timescales, susceptible depletion and replenishment drive population-scale disease dynamics. Changes in the size of susceptible population change the proportion of contacts that lead to successful disease transmission. This drives differences in expected cases between different cohorts of infectors. Many deterministic epidemiological models also include *explicit heterogeneity*, for example by adding compartments for different demographic or geographic groups [17], which can have different average rates of mixing, transmission, or recovery. When implemented in terms of individual events, the inclusion of explicit heterogeneity reproduces superspreading in linear compartmental models [14]. Despite shared theory motivating models that

highlight individual-level transmission, and those that emphasize population-level transmission dynamics, a gap persists in how each model framework treats individual-level transmission heterogeneity.

The canonical SIR (Susceptible-Infected-Removed) model uses ordinary differential equations (ODEs) to describe disease dynamics. It assumes constant, uniform infectivity and recovery times across all individuals during the infectious stage [8, 10]. Although it appears homogeneous, even the canonical SIR model has two sources of *emergent heterogeneity.* The changing effective reproductive number arising from susceptible dynamics drives variation between fractions of the population infected at different times. Moreover, the exponentially distributed recovery times assumed by the ODE drive variation in the number of infections attributable to fractions of the population infected at a given time. In other words, even the fully deterministic SIR model predicts variation in the expected number of secondary cases.

Here we delineate sources of individual variation in transmission inscribed in simple dynamical models. We trace how these sources compose into observed or observable heterogeneity in different settings, for example contrasting how heterogeneity emerges in dynamical models versus linearized, or branching-process models. We begin by revisiting early findings that intrinsic heterogeneity (e.g., due to differences in the basic reproduction number, $\mathcal{R}_0$) drives differences in variation in the secondary case distribution. However, when accounting for the feedback of transmission on disease dynamics, we find via simulations and an analytical proof that emergent heterogeneity is an invariant across a full SIR outbreak and independent of differences in intrinsic heterogeneity. As a result, expected differences in superspreading may be a consequence, in part, of focusing on early rather than the full scope of outbreaks. Throughout, we highlight the understudied but critically important role of disease dynamics *per se* in generating individual-level heterogeneity. We aim to spur the development of theory and methods that better integrate heterogeneity and dynamics, ultimately facilitating feedback between data on incidence and individual-level transmission chains, mechanistic model specification, identification, validation, and intervention design.

2 Results

2.1 Heterogeneity in the secondary case distribution in the linearized SIR model

[**JD:** *I'm voting for distinguishing our distributions primarily by the words expected vs. realized (see below).*] The early outbreak phase of an epidemic is modeled using a linearized SIR model, which assumes that the pool of susceptible individuals does not change appreciably. Early dynamics can thus be characterized by a transmission rate and an infectious period. Demographic stochasticity generates heterogeneity in this model despite the lack of explicit accounting of individual-level variation in transmission rates and infectious peri-

ods. Since recovery is Markovian, the model implicitly assumes an exponentially distributed infectious period, resulting in an exponentially distributed expected infectiousness (or *individual reproductive number*) with mean $\mu = \mathcal{R}_0$ and dispersion $\kappa = 1$. The resulting distribution of secondary cases is geometric and has a mean equal to $\mathcal{R}_0$ (see Box 1). Thus, the expected and realized secondary-case distributions are characterized by $\mathcal{R}_0$ (Figure 1a). [**JD:** *Flagging that I cut a lot of stuff around here, see if you think anything important is missing.*]

The heterogeneity in the secondary case distribution means that a large fraction of the initial secondary cases are generated by only a few infectors, even when starting from the canonical homogeneous assumption. The proportion of *expected* cases caused by the individuals who remain infectious the longest is the same for any value of $\mathcal{R}_0$, but the proportion or *realized* cases caused by the most effective infectors is larger for smaller values of $\mathcal{R}_0$, since the relative importance of the Poisson infection process increases (Figure 1b). As $\mathcal{R}_0$ becomes smaller, an increasing fraction of infected individuals generate no secondary infections, so the individuals who do transmit make up a smaller fraction of all infectors, such that realized heterogeneity is expected to increase as $\mathcal{R}_0$ decreases.

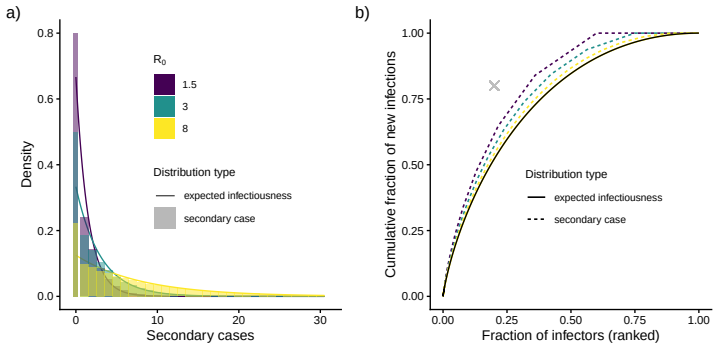


Figure 1: **Heterogeneity in the linearized SIR model** due to implicit variation in recovery times among infectors. (a) Both expected infectiousness (density curves) and secondary case distributions (density histograms) for the outset of an SIR epidemic, where the population is fully susceptible. (b) Inequality curves, defined as the cumulative fraction of new infections versus the cumulative fraction of infectors. Curves (in black) for the expected infectiousness across varying $\mathcal{R}_0$ coincide. Inequality curves derived from the secondary case distributions shown with dashed lines. The cross marks "20-80" point, where 20% of infectors cause 80% of infections. In panel a, because the first bin (at zero) sits at the boundary of support for each distribution, we have plotted this bin as double the density and half the width; this adjustment preserves area-to-area correspondence with the probability density function, while facilitating visual comparison of the heights of the density and mass functions.

2.2 Heterogeneity in secondary cases over an epidemic looks independent of $\mathcal{R}_0$

Next we consider the impact of depletion of susceptible population during outbreaks on secondary case distributions. We simulated stochastic SIR epidemics with different $\mathcal{R}_0$ values in a population of 10000 individuals. For each infected individual, we recorded the number of secondary cases they generated during the outbreak. Figure 2 shows realized distributions of secondary cases caused by individual infectors over the entire outbreaks. Unlike in the linearized scenario shown in Figure 1, where the susceptible pool remains fully populated, the secondary case distributions, computed across the entire outbreak, appear indistinguishable across a broad range of the key parameter $\mathcal{R}_0$ (compare panels (a) of Figure 1 and Figure 2).

[**JD:** *Talking about the higher variation driven by small $\mathcal{R}_0$ feels like a red herring here and I cut it out. The key point is that we expected within-cohort κ to be the same, and between-cohort κ to increase with $\mathcal{R}_0$. Is it time to talk about cohorts here?*][**TG:** *Not yet. Hard to introduce cohorts in the context of the individual-based SIR. We dont introduce cohorts until the next section.*]

Two key features of the secondary case distributions across an entiure outbreak are salient. First, the mean of the secondary case distribution is approximately 1, as expected. Since we are considering a single outbreak seeded by single initial case, the number of infectors must almost equal the number of infectees. Second, the variance of the realized secondary case distribution is approximately 2, corresponding to a variance of 1 in the expected case distribution, independent of $\mathcal{R}_0$. This result is unexpected. In the absence of susceptible depletion, heterogeneity in expected case number is constant across values of $\mathcal{R}_0$ (Figure 1b). During the full epidemic, we expect large-$\mathcal{R}_0$ simulations to cause large dynamical changes (Figure 3b), leading to larger variation in expected (and thus realized) case numbers. At first glance, it is not clear what could balance the dynamical variation in susceptibles for different values of $\mathcal{R}_0$ to leave the variance constant.

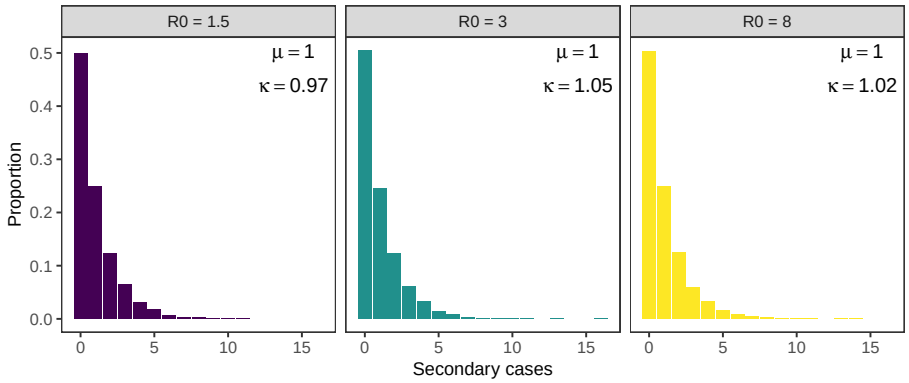


Figure 2: **Heterogeneity in the number of secondary cases over an entire epidemic is independent of $\mathcal{R}_0$.** Realized secondary cases collected from individual-based simulations of the SIR model for $\mathcal{R}_0 = 1.5, 3$, and 8 (from left to right respectively); μ and κ denote the mean and dispersion of the realized case distribution obtained from a single realization, conditional on a large outbreak occurring, of the individual-based model for each $\mathcal{R}_0$, in a population of 10^4 individuals with one initial infectious individual.

Box 1. Linking variance across discrete and continuous time models of epidemic outbreaks

During the early stages of an outbreak, secondary-cases counts are typically modeled using a negative-binomial distribution, with mean μ and shape, or "dispersion," parameter k. This distribution is linked through Poissonification, i.e., drawing individual infection events randomly, to a Gamma distribution with the same mean and shape parameter values. Here, the Gamma distribution describes variation among individuals in expected infectiousness.

We note a couple of points. First, the squared coefficient of variation (SCV) of the underlying Gamma distribution is $\kappa = 1/k$, so its variance is $\sigma^2_{\text{act}} = \kappa\mu^2$. The resulting variance in secondary-case counts is $\sigma^2 = \kappa\mu^2 + \mu$, with the last term arising from the Poisson step of the negative-binomial construction. Second, the relationship between σ^2_{act} and σ^2 holds regardless of the form of the expected infectiousness distribution [11].

As κ approaches zero, the distribution of expected infectiousness converges to a point mass at μ, while the corresponding distribution of secondary-case counts converges to a Poisson distribution with mean μ. We therefore use κ, as a (generalized) *dispersion* parameter, with larger values indicating greater heterogeneity in expected infectiousness and secondary cases.

[**JD:** *We need to be very careful about the word cohort (and I am going to*

run out of time for the night). I am thinking that we should call the two-ended cohorts "interval cohorts" and the one-ended cohorts "forward cohorts". Will try to add a few examples.]

2.3 Heterogeneity in secondary cases over an epidemic is independent of $\mathcal{R}_0$ – Proof

We set out to explore the basis for the observation that both the mean and variance of realized cases in the stochastically realized SIR model appear to be invariant with respect to changes in $\mathcal{R}_0$. To do so, consider the deterministic ODE-based SIR model:

$$\begin{aligned} \frac{dx}{dt} &= -\mathcal{R}_0 x(t) y(t), \\ \frac{dy}{dt} &= \mathcal{R}_0 x(t) y(t) - y(t), \end{aligned} \tag{1}$$

where $x(t)$ is the proportion of susceptible individuals at time t, $y(t)$ is the proportion of infectious individuals at time t, t is time measured in units of the mean infectious duration and $\mathcal{R}_0$ is the basic reproduction number.

To calculate the mean and variance of the secondary cases, we must track infections and the infectors that caused those infections. While individual infectors and cases cannot be tracked in this deterministic SIR model, we can track cohorts of infectors and the cases caused by them. By calculating the weighted moments of the expected cases per case caused by each cohort, we can obtain the heterogeneity of the secondary cases per case over the epidemic.

We define an "interval cohort", c_{τ_1,τ_2}, as the fraction of infectors that become infectious at time τ_1 and stop being infectious at time τ_2. The weight of the cohort, $w(\tau_1, \tau_2)$ is the number of infectors in the cohort. The number of infectors who become infectious at time τ_1 is $i(\tau_1) = \mathcal{R}_0 x(\tau_1) y(\tau_1)$. This quantity, $i(\tau_1)$ is also called the incidence of infection, or the number of new cases that appear at time τ_1. Only a fraction of these infectors remain infectious till time τ_2. If $f(\cdot)$ denotes the probability density function of the infectious period, then

$$w(\tau_1, \tau_2) = i(\tau_1) f(\tau_2 - \tau_1) \tag{2}$$

Now that we have the cohort weights, we need to calculate the number of cases per case caused by the cohort. The total number of cases caused by the cohort c_{τ_1,τ_2} over the course of the epidemic is

$$\hat{C}(\tau_1, \tau_2) = \int_{\tau_1}^{\tau_2} \mathcal{R}_0 x(t') w(\tau_1, \tau_2) dt' = \mathcal{R}_0 w(\tau_1, \tau_2) \int_{\tau_1}^{\tau_2} x(t') dt' \tag{3}$$

Then the expected cases per case caused by cohort c_{τ_1,τ_2} is

$$C(\tau_1, \tau_2) = \frac{\hat{C}(\tau_1, \tau_2)}{w(\tau_1, \tau_2)} = \mathcal{R}_0 \int_{\tau_1}^{\tau_2} x(t') dt' \tag{4}$$

Now that we have the number of expected cases per case and weight for each cohort, we can calculate the moments of the distribution of expected secondary cases. The kth weighted raw moment of this distribution is

$$C_k = \iint\limits_{0<\tau_1\leq\tau_2<\infty} d\tau_1 d\tau_2 \; w(\tau_1,\tau_2)(C(\tau_1,\tau_2))^k, \tag{5}$$

With this definition of the raw moments, the assertion that the heterogeneity of the secondary case per case distribution over the entire epidemic is invariant is equivalent to:

$$\kappa = \frac{C_0 C_2}{C_1^2} - 1 = 1 \tag{6}$$

[**JD:** *Took out the intuition for $C_1 = C_0$. We already have it above, and it will disappoint some people who will start off think we're going to get to the interesting part of the result.*] We calculate the moments and confirm the result in section S1.1. We note that the result follows from two assumptions: the distribution of the infectious duration is memoryless, and the initial infectious cohort is infinitesimal, i.e., $y(t=0) \to 0$. The first assumption is true for the canonical SIR model where the duration of infectious periods is exponentially distributed. We discuss relaxing the second assumption in section S1.2.

2.4 Balance between within- and between-cohort components of the variance in secondary cases illuminates how heterogeneity in secondary cases is independent of $\mathcal{R}_0$

To understand why the heterogeneity in secondary cases over the course of the epidemic is independent of $\mathcal{R}_0$, we divide the population into "forward cohorts" of individuals infected at the same time, and use variance partitioning to split the heterogeneity into two components: a within-cohort component and a between-cohort component. Individuals within a cohort have access to the same susceptible pool initially but have differences in their secondary case counts as a result of their exponentially distributed infectious periods. Individuals in different forward cohorts also differ in the starting size of the susceptible pools that they have access to. It is important to note that both early and late cohorts have the same distributions of infectious periods and expected infectiousness but due to the depletion of the susceptible pool over time (Figure 3b), later cohorts, on average, have much lower secondary cases than earlier cohorts. Thus, the between-cohort variance can be attributed to susceptible depletion.

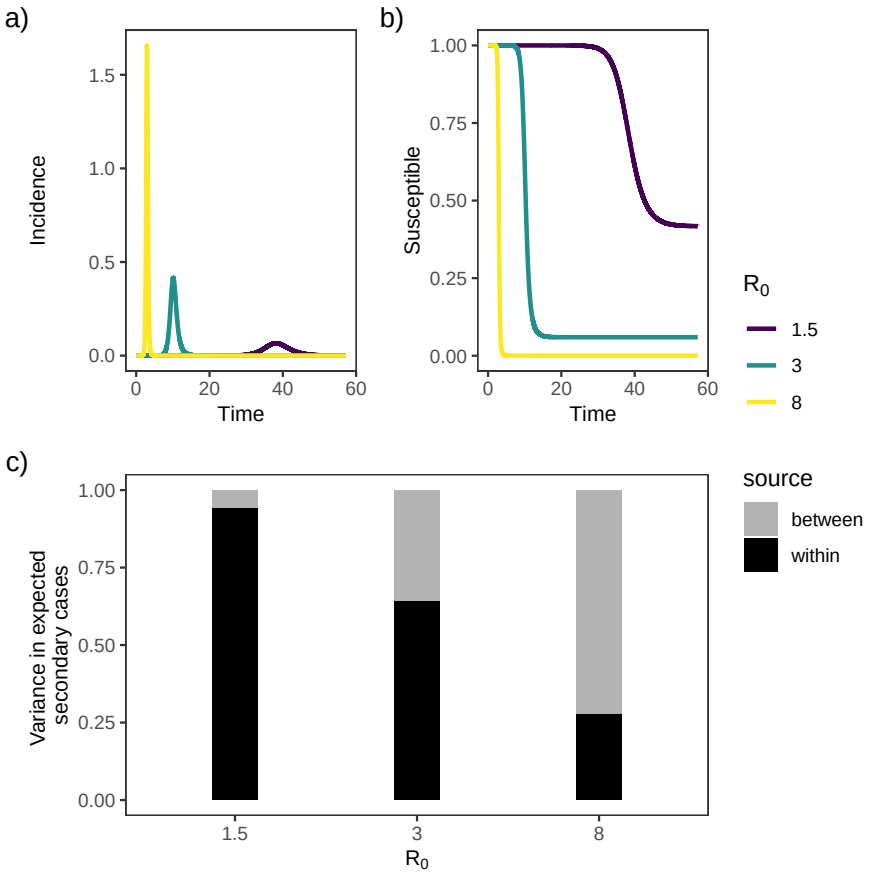


Figure 3: **Deterministic SIR dynamics and within- and between-cohort variance** (a) Incidence and (b) proportion of susceptibles as functions of time, for different values of $\mathcal{R}_0$. The epidemics were modeled using an ODE-based SIR model. Time is in units of mean infectious period.(c) Stacked bar plot showing within- and between-cohort components of variance in secondary cases calculated numerically for different values of $\mathcal{R}_0$.

Numerical calculation of the within- and between-cohort components of the variance in secondary cases using the ODE-based SIR model (see methods for details of the calculation) shows that as $\mathcal{R}_0$ increases, the between-cohort component of the variance increases and the within-cohort component of the variances decreases (Figure 3c). However, the sum of the two components, remains constant across $\mathcal{R}_0$. As $\mathcal{R}_0$ increases, the variance in the size of infectious cohorts (Figure 4a) and well as the speed and severity of susceptible depletion also increases (Figure 3b). As a result of the susceptible depletion, the mean number of secondary cases also decreases rapidly across cohorts (Figure 4b),

with earlier cohorts having a much higher mean of secondary cases compared to later cohorts. Further, the decline in the mean number of secondary cases across cohorts becomes sharper as $\mathcal{R}_0$ increases. As a result, the between-cohort variance in the number of secondary cases increases with $\mathcal{R}_0$.

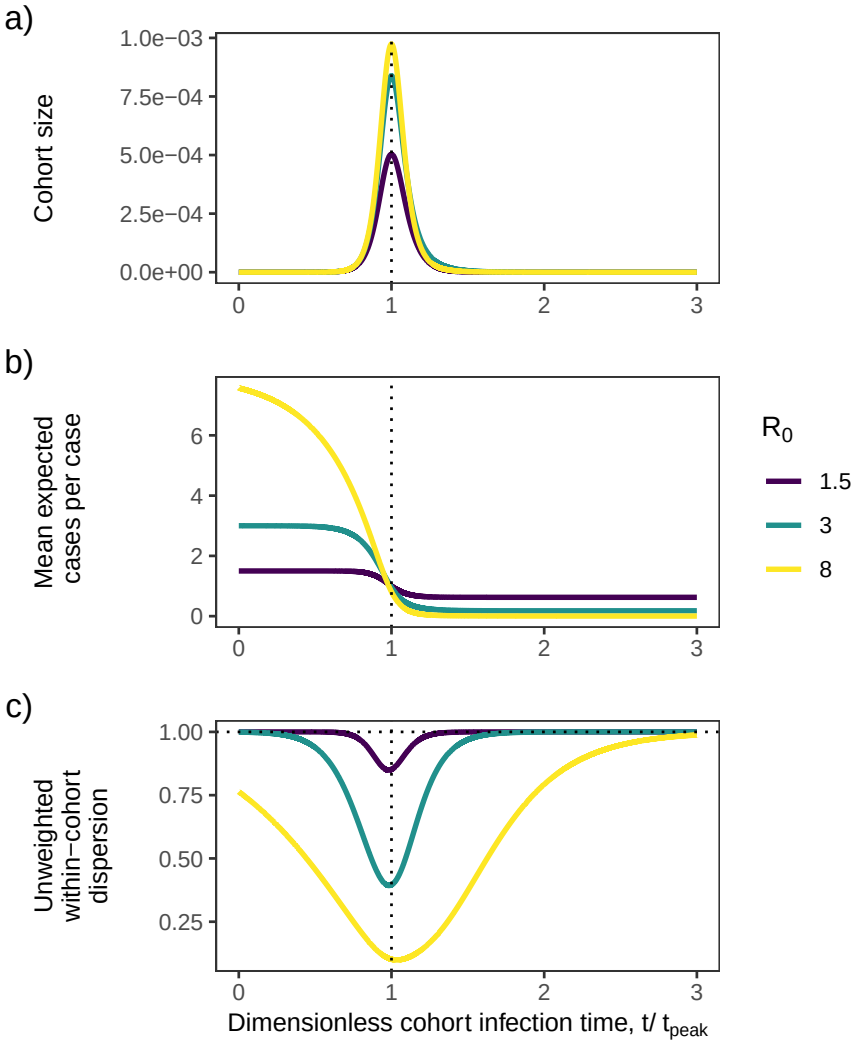


Figure 4: **Expected infectiousness of cohorts.** (a) Cohort size as a function of time of infection. Time has been normalized by the time of peak incidence (t_{peak}). Individuals infected during $[t, t + \Delta t]$ form a cohort, with $\Delta t = 2 \times 10^{-4} t_{\text{peak}}$. (b) Mean expected cases per case for each cohort. (c) Within-cohort dispersion of secondary cases. Note that the x-axes refer to cohort infection time. For example, the mean expected cases per case at $t/t_{\text{peak}} = 1$ does not refer to the mean expected cases per case at time $t = t_{\text{peak}}$. Instead, it refers to the mean expected cases per case of the cohort that became infectious at time $t = t_{\text{peak}}$.

The within-cohort variance, on the other hand, decreases as $\mathcal{R}_0$ increases (Figure 3c). Since the mean number of secondary cases varies widely between early and late cohorts, we look at the within-cohort dispersion in secondary cases to better understand how within-cohort heterogeneity changes between early vs late cohorts for an epidemic with a given $\mathcal{R}_0$. For a given $\mathcal{R}_0$, early cohorts have high dispersion in secondary cases, driven by access to a large susceptible pool (Figure 4c). As the epidemic progresses, the susceptible pool depletes (Figure 3b) and the within-cohort dispersion in secondary cases decreases (Figure 4c). The within-cohort dispersion increases for the later cohorts, driven by the shrinking within-cohort mean in the number of secondary cases (Figure 4c). However, the within-cohort contribution to the total variance depends not just on the within-cohort dispersion, but also on the cohort weight. Once the contributions of each cohort to the variance are appropriately weighted by the cohort mean and cohort size, we find that the weighted within-cohort dispersion is low for both early and late cohorts, and is high for cohorts that arise in the middle of the epidemic (Figure 4a) for a given $\mathcal{R}_0$. Indeed, across different values of $\mathcal{R}_0$, we find that the weighted within-cohort dispersion decreases as $\mathcal{R}_0$ increases (Figure 3c).

Together, the decrease in the within-cohort variances balances the increase in the between-cohort variance, leading to the total variances in secondary cases being independent of $\mathcal{R}_0$. This change in the within- and between-cohort variances is driven by susceptible depletion.

3 Discussion

[**TG:** *From meeting notes: Focus on the bridge between individual based- vs population level perspectives. Even without explicit heterogeneity, SIR model has emergent heterogeneity due to disease dynamics. The invariance result is a special case resulting from the Markovian infection process. The result does not hold for other infectious period distributions but establishes a baseline for comparison. More work is needed.*]

Quantifying heterogeneity in secondary-case distributions helps better understand how epidemics unfold and how they can be contained. Although there have been efforts to quantitatively describe individual-level heterogeneity in contact rates [19] and to parameterize biologically realistic secondary case distributions, abstracted from the mechanism [13], these efforts have been largely separate from the construction and evaluation of the compartmental models that enable epidemic forecasting and intervention design (but see [14]). Here, we expand the bridge between individual- and population perspectives of heterogeneity by studying epidemiologically-relevant cohorts of infectors that become infectious at the same time.

We first show how demographic stochasticity and a Poisson contact process result in heterogeneity in secondary cases, even in the linearized SIR model with its canonical assumption of homogeneity between individuals, and lack of susceptible depletion. Next, we show how heterogeneity emerges in the SIR model of disease dynamics. Depletion of susceptibles over the course of an epidemic leads to differences in realized secondary cases between early and late infectors, despite sharing identical underlying infectiousness distributions. We prove a surprising result – that the total variance in secondary cases over an epidemic is constant across all values of $\mathcal{R}_0$. This invariance is in contrast with the $\mathcal{R}_0$-dependent variance of secondary cases in the linearized SIR model. We show that the invariance arises due to a balance between the increase in between-cohort variance and the decrease in within-cohort variance, both driven by the dynamics of susceptible depletion.

Our cohort-level analyses reveal the importance of susceptible depletion, even in early stages of the epidemic. We find that even the earliest cohorts of infectors have mean secondary case counts much below the theoretical $\mathcal{R}_0$ (Figure 4b). This discrepency becomes larger as $\mathcal{R}_0$ increases. In practice, this means that if surveillance begins after the initial transmission generations, mean secondary case counts will systematically underestimate $\mathcal{R}_0$ [16].

The independence of the variance in secondary cases hinges upon the memorylessness of the infectious period distribution. This means that the result only holds for exponentially or geometically distributed infectious periods, which is only true in the SIR model. Further, the SIR model assumes a closed, well-mixed population. It is therefore, also unclear how heterogeneity in mixing propensities might impact our results. However, our results with the SIR model provide a baseline for comparing heterogeneity in secondary cases across different models. The results also reveal how the dispersion parameter, κ, can be used as a unified metric of heterogeneity across individual-based and compartmental

models that we hope is of service in extending the current results to a richer set of model assumptions.

4 Materials and Methods

4.1 Individual-based SIR model

We simulate an individual-based SIR model for a population of N individuals. Each individual is assigned a numerical index and an intrinsic recovery time, drawn from an exponential distribution with a mean recovery time $\frac{1}{\gamma}$. Next, we build a matrix containing all possible pairs of individuals, and generate a pool of contact events by randomly sampling pairs from this matrix with replacement. The contact rate for each pair of individuals is $\beta/(N-1)$, where β is the transmission rate. The total rate of contact events in the population is obtained by multiplying this rate by the total number of possible pairs, $N(N-1)/2$, giving $\beta N/2$. The time interval between successive contact events is then drawn from an exponential distribution with a rate $\beta N/2$,

Throughout the simulation, we track the infection status and number of infections caused by each individual. We initialize the simulation by assigning a single random individual to be infectious. We then run the simulation forward in time, with successive time steps corresponding to successive contact events. At each timestep, we assess the infection status of each individual based on when they became infectious and their assigned recovery time. Then, we check the infection statuses of the individuals that come in contact: if both individuals are susceptible or infectious or recovered/removed, we move to the next timestep. If one of the individuals is recovered/removed, we move to the next timestep. However, if one of the individuals is infectious and the other is susceptible, we change the infection status of the susceptible individual to infectious, noting the time when they become infectious. We also increase the tally of secondary cases caused by the infectious individual by 1. Then, we move to the next timestep. We stop this process when either no infectious or no susceptible individuals remain.

4.2 Deterministic SIR model

The deterministic ODE-based SIR model is:

$$\begin{aligned} \frac{dS}{dt} &= -\beta \frac{SI}{N}, \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I, \\ \frac{dR}{dt} &= \gamma I, \end{aligned} \tag{7}$$

where $S(t)$, $I(t)$ and $R(t)$ are the number of susceptible, infectious and recovered/removed individuals at time t respectively. $N = S + I + R$ is the total

number of individuals. β is the transmission rate and γ is the recovery rate. We do not need to track R explicity, since $R = N - S - I$ and N is a constant ($\frac{dN}{dt} = 0$). We rewrite the equations in terms of the rescaled variables, $x = \frac{S}{N}$, $y = \frac{I}{N}$ and $\tau = \gamma t$, to get:

$$\begin{aligned} \frac{dx}{d\tau} &= -\mathcal{R}_0 xy, \\ \frac{dy}{d\tau} &= \mathcal{R}_0 xy - y, \end{aligned} \tag{8}$$

where $\mathcal{R}_0 = \frac{\beta}{\gamma}$ is the basic reproduction number. $x(\tau)$ is the proportion of susceptible individuals, $y(\tau)$ is the proportion of infectious individuals and τ is time in units of the mean recovery period $\frac{1}{\gamma}$.

4.3 Calculation of cohort mean and within-cohort variance of secondary cases

The values of secondary cases in panel (a) of Figure 1 were generated by computing the geometric probability mass function with mean $\mathcal{R}_0$ at points $0, 1, \ldots, 30$. We used the convention that the geometric distribution counts the number of failures before the first success. As for the expected infectiousness, we evaluated the exponential probability density function with mean $\mathcal{R}_0$ at 300 equally distanced point in the interval $[0, 30]$.

To generate panel (b) of Figure 1, we followed Lloyd-Smith's approach and computed the inequality in the activity distribution [13, SI, Sec. 2.2.5]. More specifically, we used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0^2} \int_0^x v e^{-\frac{v}{\mathcal{R}_0}} dv$ which is the fraction of cases due to those caused up to x secondary cases. The fraction of cases due to those caused more than x cases would be $1 - F_{\text{trans}}(x)$, which is equal to $(1 + \frac{x}{\mathcal{R}_0}) e^{-\frac{x}{\mathcal{R}_0}}$. The population fraction of the individuals infected more than x cases is $e^{-\frac{x}{\mathcal{R}_0}}$. We used a similar approach to compute the inequality in the secondary case distribution: We used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0} \sum_{v=0}^{x} v G(v, \mathcal{R}_0)$ which is the fraction of cases due to those caused up to x secondary cases. Here $G(v, \mathcal{R}_0)$ is a geometric distribution with mean $\mathcal{R}_0$. The fraction of cases due to those caused more than x cases equals $1 - F_{\text{trans}}(x)$. The population fraction of these individuals would be $1 - P(x)$, where $P(x)$ is the cumulative distribution function of a geometric distribution with mean $\mathcal{R}_0$ evaluated at x.

To generate Figure 2, we used the individual-based SIR model as detailed in Section 4.1 and computed the number of secondary cases in a population of size 10^4.

We used a simple deterministic SIR model to compute the mean and variance of expected secondary cases C. First, we scaled time by the mean infectious period, so the resulting SIR model then depends on one parameter, $\mathcal{R}_0$. The SIR differential equations were numerically solved for the proportions of susceptible $x(t)$ and infectious $y(t)$ at each time point t. We then used the `R` function `approxfun` to construct a function, $X(t)$, that linearly interpolates the time-series $(t, \mathcal{R}_0 x(t))$. We associated a (forward) cohort to each of the first 60%

time points in the time-series t (We used a 60% cutoff to ensure cohorts had almost recovered by the end of the simulation period) and calculated the cohort-specific mean and variance.

More specifically, for the cohort infected at time point τ, the mean, $\mu(\tau)$, and variance $\sigma^2(\tau)$ of expected secondary cases read as:

$$\mu(\tau) = \int_{t>\tau} f(t-\tau)C(\tau,t)dt\,, \tag{9a}$$

$$\mathbb{E}[C^2(\tau)] = \int_{t>\tau} f(t-\tau)C^2(\tau,t)dt\,, \tag{9b}$$

$$\sigma^2(\tau) = \mathbb{E}[C^2(\tau)] - \mu^2(\tau). \tag{9c}$$

Here $f(\cdot)$ is the probability density function of the infectious period. To calculate these quantities for the forward cohort infected at τ, the interpolated $\mathcal{R}_0 x(t)$ was integrated from τ to t. This gives us the expected total number of cases caused by the cohort fraction recovered at time point t, i.e., $C(\tau,t)$. The mean expected case per case for each cohort $\mu(\tau)$ was then the integral of interval cohorts' expected total numbers of cases weighted by the infectious period density function. We used the same approach to compute $\mathbb{E}[C^2(\tau)]$ and variance of C for each cohort ($\sigma^2(\tau)$). In Figure 4, the middle panel (panel b) was generated using this approach.

We obtained mean expected secondary cases C by integrating the cohort-specific mean $\mu(\tau)$ against the incidence, $i(\tau) = \mathcal{R}_0 x(\tau) y(\tau)$, and normalizing the result by the final size, $\int i(t)dt$. To compute the between-cohort component of variance, we integrated the squared cohort mean $\mu^2(\tau)$ weighted by the incidence $i(\tau)$ and divided it by the final size. The result minus the squared mean of C gave us the between-cohort component of variance. The within-cohort component of the variance was computed by taking an integral over the cohorts' variance $\sigma^2(\tau)$ weighted by the incidence. The result was also normalized by the final size. The total variance was the sum of the between- and within-cohort components of variance. We also calculated the total variance independently; we integrated $\mathbb{E}[C^2(\tau)]$ weighted by the incidence and divided it by the final size. The result minus the squared mean of C yielded the total variance. Both approaches produced the same value for the total variance.

[**AA:** *removed simulations details for the supp figures; will put them back if we decide to keep those figures*]

All integrations were done using `lsoda` method with the `R` package `deSolve`. All simulations were carried out using `R 4.5.2`. Code for all numerical simulations is housed at: `https://github.com/dushoff/kappaCode`. We solved all integrals across a range of values for $\mathcal{R}_0$, using the starting values $y_0 = 10^{-9}; x_0 = 1 - 10^{-9}$ to represent the limiting case in which there are no exogenous cases.

[**AA:** *What about taking this part out?*] In building these simulations, we used a range of time step sizes, noting convergence towards known and conjectured values (e.g., epidemic final size, mean case reproduction number, variance in case reproduction number) as resolution increased.

Declaration of competing interests

No competing interests declared.

Data and code availability

All code and simulation files can be found at https://github.com/dushoff/kappaCode and have been archived via Zenodo [X].

Acknowledgments

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S.1 Analytical Results

The standard SIR model is described by the following equations:

$$\dot{x}(t) = -\mathcal{R}_0 x(t) y(t),$$
$$\dot{y}(t) = \mathcal{R}_0 x(t) y(t) - y(t),$$

where $x(t)$ and $y(t)$ denote the proportion of susceptible and infectious individuals, respectively. Time has been normalized by the recovery rate. Let $i(t)$ denote the incidence $\mathcal{R}_0 x(t) y(t)$, and define the size of the epidemic as follows:

$$Z = \int_0^\infty dt i(t).$$

Let $f(\cdot)$ denote the probability density function of the infectious period. Define $F(t)$ as the corresponding survival distribution, that is

$$F(t) = 1 - \int_0^t d\tau f(\tau) = \int_t^\infty d\tau f(\tau). \tag{S.1}$$

Note that $F(t)$ is the probability that an individual infected at time 0 remains infectious after time t.

We define the *expected case reproductive number* between time τ and ρ as

$$C(\tau, \rho) = \mathcal{R}_0 \int_\tau^\rho dt\ x(t).$$

The kth weighted raw moment of C is then defined as

$$C_k = \int_0^\infty d\tau \int_\tau^\infty d\rho\ w(\tau, \rho)(C(\tau, \rho))^k, \tag{S.2}$$

where $w(\tau, \rho) = i(\tau) f(\rho - \tau)$ is the size of the cohort that became infectious at time τ and recovered at time ρ.

In Section S.1.1, we prove the following statement: The excess variation of the individual case reproductive number in the standard SIR model outbreak is 1, provided that the initially infectious cohort size is negligible. Mathematically, this amounts to proving that

$$\kappa = \frac{C_0 C_2}{C_1^2} - 1 = 1. \tag{S.3}$$

S.1.1 Proof

We set out to calculate C_0, C_1 and C_2. We can calculate C_0 and C_1 without making any assumptions about the functional form of f (we do not have to assume that the wait times are exponentially distributed). We do assume that F is memoryless, i.e., obey $F(a + b) = F(a)F(b)$. Calculating each of these

moments involves multiple manipulations of the limits of the integrals. To proceed, we assume that size of the initial infectious cohort is negligible.

From Equation S.2, we have

$$\begin{aligned} C_0 &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; w(\tau, \rho) \\ &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; i(\tau) f(\rho - \tau) \end{aligned}$$

Substituting $\omega + \tau$ for ρ in the inner integral, results in

$$C_0 = \int_0^\infty d\tau \; i(\tau) \int_0^\infty d\omega \; f(\omega).$$

Since f is the residence time distribution, the inner integral is equal to 1. It follows that

$$\begin{aligned} C_0 &= \int_0^\infty d\tau \; i(\tau) \\ &= Z. \end{aligned}$$

The next step is to prove that $C_1 = Z$. From Equation S.2, we have

$$\begin{aligned} C_1 &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; w(\tau, \rho) C(\tau, \rho) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty d\rho i(\tau) f(\rho - \tau) \int_\tau^\rho dt x(t). \end{aligned}$$

Instead of first integrating over t from τ to ρ and then over ρ from τ to ∞, we can first integrate over ρ from t to ∞ and then integrate over t from τ to ∞, i.e.,

$$C_1 = \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau) x(t) \int_t^\infty d\rho f(\rho - \tau).$$

Substitute $\rho = \omega + \tau$ in the innermost integral and using Equation S.1, we get

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt \; i(\tau) x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt \; i(\tau) x(t) F(t - \tau). \end{aligned}$$

Note again that we have not used the fact that f is exponential, we have merely used the definition of F as an integral of f. By interchanging the order of the integrals once more, we have

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt \; i(\tau) x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty dt \; x(t) \int_0^t d\tau \; i(\tau) F(t - \tau). \end{aligned}$$

Note that the inner integrand counts the number of individuals that entered the infectious state at some time $\tau < t$ and remain infectious at time t. When we integrate over all times $\tau < t$, we get the number of individuals who are infections at time t, i.e., $y(t)$. So,

$$C_1 = \mathcal{R}_0 \int_0^\infty dt\ x(t) \int_0^t d\tau\ i(\tau) F(t-\tau) \tag{S.4}$$

$$= \mathcal{R}_0 \int_0^\infty dt\ x(t) y(t) \tag{S.5}$$

$$= \int_0^\infty dt\ i(t) \tag{S.6}$$

$$= Z. \tag{S.7}$$

The next step is to show $C_2 = 2Z$. Given Equation S.2, we have

$$C_2 = \int_0^\infty d\tau \int_\tau^\infty d\rho\ w(\tau,\rho) \left(C(\tau,\rho)\right)^2$$

$$= \mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho\ i(\tau) f(\rho-\tau) \left(\int_\tau^\rho dt x(t) \right)^2$$

Now we break down the square integral into a double integral

$$\left(\int_\tau^\rho x(t) dt \right)^2 = \left(\int_\tau^\rho x(t) dt \right) \left(\int_\tau^\rho x(s) ds \right)$$

$$= \left(\int_\tau^\rho x(t) dt \right) \left(\int_\tau^\rho x(s) ds \right)$$

$$= \int_\tau^\rho \int_\tau^\rho dt ds\ x(t) x(s).$$

In $t-s$ space, this is integrating $x(t)x(s)$ over a square region whose vertices are (τ,τ), (τ,ρ), (ρ,τ) and (ρ,ρ). This is twice the integral over the triangle whose vertices are (τ,τ), (τ,ρ) and (ρ,τ). So,

$$\left(\int_\tau^\rho x(t) dt \right)^2 = 2 \int_\tau^\rho ds \int_\tau^s dt\ x(t) x(s)$$

Plugging this back into the expression for C_2 yields

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho \int_\tau^\rho ds \int_\tau^s dt\ i(\tau) f(\rho-\tau) x(t) x(s).$$

First, we reorder the integrals over ρ and s and change the limits appropriately

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_s^\infty d\rho \int_\tau^s dt f(\rho-\tau) x(t).$$

Next, we reorder the integrals over ρ and t to get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_\tau^s dt\ x(t) \int_s^\infty d\rho f(\rho - \tau).$$

By substituting $\rho = \omega + \tau$ and using Equation S.1, we get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_\tau^s dt\ x(t) F(s - \tau).$$

We then reorder the integrals over τ and s

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s d\tau \int_\tau^s dt\ x(t) i(\tau) F(s - \tau).$$

Swapping the integrals over τ and t, results in

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t) \int_0^t d\tau\ i(\tau) F(s - \tau).$$

Now, we use the fact that the survival function F is memoryless, i.e., $F(a+b) = F(a)F(b)$, and replace $F(s-\tau)$ with $F(s-t)F(t-\tau)$

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t) \int_0^t d\tau\ i(\tau) F(s - t) F(t - \tau).$$

Next, we use the same argument we used to go from Equation S.4 to Equation S.5, which results in

$$\begin{aligned}
C_2 &= 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t) F(s - t) y(t) \\
&= 2\mathcal{R}_0 \int_0^\infty ds\ x(s) \int_0^s dt\ i(t) F(s - t) \\
&= 2\mathcal{R}_0 \int_0^\infty ds\ x(s) y(s) \\
&= 2 \int_0^\infty ds\ i(s) \\
&= 2Z.
\end{aligned}$$

Finally, replacing C_0, C_1, and C_2 by Z, Z, and $2Z$, respectively, in Equation S.3, results $\kappa = 1$.

S.1.2 Incorporating the Size of Initial Infections

Equation S.3 was based on the assumption that the population who were infectious initially was negligible. In what follows, we revisit the results while relaxing this assumption. To account for the initial infections, we assume that the epidemic starts abruptly at time $t = 0$, i.e.,

$$i(t) = y_0 \delta(t), \quad t \in (-\infty, 0]. \tag{S.8}$$

Here, y_0 denotes the proportion of the population that were infectious at the beginning of the outbreak, $t = 0$, and $\delta(t)$ denotes the Dirac delta distribution [2]. Then, accounting for the initially infectious cohort, the proportion of the infected population that is infectious is given by

$$y(t) = \int_{0^-}^{t} d\tau \; i(\tau) F(t-\tau).$$

Here, the lower limit of the integral is 0^- to include the Dirac delta function at $t = 0$. The total weight C_0 will now includes the initially infectious cohort size as well, i.e,

$$C_0 = \int_{0^-}^{\infty} dt \; i(t) \tag{S.9}$$

The first raw moment C_1 is given by

$$\begin{aligned}
C_1 &= \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} d\rho \; w(\tau,\rho) C(\tau,\rho) \\
&= \mathcal{R}_0 \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} dt i(\tau) x(t) \int_{t}^{\infty} d\rho f(\rho-\tau) \\
&= \mathcal{R}_0 \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} dt i(\tau) x(t) F(t-\tau) \\
&= \mathcal{R}_0 \int_{0}^{\infty} dt x(t) \int_{0^-}^{t} d\tau i(\tau) F(t-\tau) \\
&= \mathcal{R}_0 \int_{0^-}^{\infty} dt \; x(t) y(t).
\end{aligned}$$

Given both $x(t)$ and $y(t)$ are bounded, without loss of generality, the lower limit can be replaced with 0^+. Thus, the first raw moment C_1 becomes

$$\begin{aligned}
C_1 &= \mathcal{R}_0 \int_{0^+}^{\infty} dt \; x(t) y(t) \\
&= \int_{0^+}^{\infty} dt \; i(t).
\end{aligned}$$

We note that C_1 does not include the size of initially infectious cohort. Now we move on to recalculate C_2, while taking the initially infectious cohort into account. We have

$$\begin{aligned}
C_2 &= \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} d\rho \; w(\tau,\rho) \left(C(\tau,\rho)\right)^2 \\
&= \mathcal{R}_0^2 \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} d\rho \; i(\tau) f(\rho-\tau) \left(\int_{\tau}^{\rho} dt x(t) \right)^2
\end{aligned}$$

Using the same mathematical manipulation as in section S.1.1, we obtain

$$\begin{aligned} C_2 &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds\; x(s) \int_{0^-}^{s} dt\; x(t) \int_{0^-}^{t} d\tau\; i(\tau) F(t-\tau) F(s-t) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds\; x(s) \int_{0^-}^{s} dt\; x(t) y(t) F(s-t) \end{aligned}$$

We split the innermost integral into two integrals

$$\begin{aligned} C_2 &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds\; x(s) \int_{0^-}^{s} dt\; x(t) y(t) F(s-t) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds\; x(s) \left(\underbrace{\int_{0^-}^{0^+} dt\; x(t) y(t) F(s-t)}_{0} + \int_{0^+}^{s} dt\; x(t) y(t) F(s-t) \right) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds\; x(s) \int_{0^+}^{s} dt\; x(t) y(t) F(s-t) \\ &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds\; x(s) \int_{0^+}^{s} dt\; i(t) F(s-t). \end{aligned}$$

The innermost integral gives the proportion of the population who entered the infectious state at some time $t > 0$ and remain infectious at time s. Hence, it equals the total proportion infectious at time s minus the proportion of the population who were initially infectious and remain infectious at time s, i.e., $y(s) - y_0 F(s)$. Having replaced the innermost integral with $y(s) - y_0 F(s)$, we obtain

$$\begin{aligned} C_2 &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds\; x(s) \left(y(s) - y_0 F(s) \right) \\ &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds\; x(s) y(s) - 2\mathcal{R}_0 y_0 \int_{0^-}^{\infty} ds\; x(s) F(s) \\ &= 2 \int_{0^+}^{\infty} i(s) ds - 2\mathcal{R}_0 y_0 \int_{0^+}^{\infty} ds\; x(s) F(s) \\ &= 2C_1 - 2\mathcal{R}_0 y_0 \int_{0^+}^{\infty} ds\; x(s) F(s). \end{aligned}$$

Note that changing the lower limit of integration from 0^- to 0^+ does not affect the result because the integrand is bounded. The second term in the left hand side is the twice the expected infectiousness of the initial cohort. Here we notice that the second raw moment is *dependent* on the key parameter $\mathcal{R}_0$. However, as the initial cohort size y_0 tends to zero, (i) the difference between C_2 and twice the first raw moment $2C_1$ approaches zero; (ii) the total weight C_0 (S.9) converges to C_1. Thus, Equation S.3 is retrieved.

S.1.3 Moments of time-truncated case reproductive number at some point during the outbreak

In this section, we derive moments of truncated C at an arbitrary time $j \in (0, \infty)$ during the outbreak, while accounting for the initially infectious cohort. Up to time j, the expected realized infections due to individuals who became infectious at time $\tau \in (0, j)$ and remain infectious at time j is the equal to that of the individuals who became infectious at τ and recovered at time j. To capture this observation, we introduce a modified infectious period density function, denoted $g_j(t, \tau)$. This function equals the true infectious period density function $f(t-\tau)$ for $t < j$, and assigns the remaining probability mass to $t = j$. Hence, its cumulative distribution function reaches 1 at j, i.e.,

$$g_j(t,\tau) = \begin{cases} f(t-\tau), & \text{if } t < j \text{ and } \tau < j, \\ F(j-\tau)\delta(t-j), & \text{if } t \geq j \text{ and } \tau < j, \\ 0, & \text{otherwise.} \end{cases}$$

Accordingly, we redefine the size of the cohort that became infectious at time τ and recovered at time ρ as follows

$$w_j(\tau,\rho) = i(\tau) g_j(\rho,\tau), \quad \tau \leq \rho \leq j.$$

Note that $\int_t^j g_j(s,\tau)\, ds$ is equal to $F(t-\tau)$ for $t < j$. It is straightforward to show that the total weight $C_0(j)$ equals

$$C_0(j) = \int_{0^-}^{j} dt\; i(t). \tag{S.10}$$

The first raw moment $C_1(j)$ is given by

$$\begin{aligned} C_1(j) &= \int_{0^-}^{j} d\tau \int_{\tau}^{j} d\rho\; w_j(\tau,\rho) C(\tau,\rho) \\ &= \mathcal{R}_0 \int_{0^-}^{j} d\tau \int_{\tau}^{j} d\rho\; i(\tau) g_j(\rho,\tau) \int_{\tau}^{\rho} dt x(t) \\ &= \mathcal{R}_0 \int_{0^-}^{j} d\tau \int_{\tau}^{j} dt\; i(\tau) x(t) \int_{t}^{j} d\rho g_j(\rho,\tau) \\ &= \mathcal{R}_0 \int_{0^-}^{j} d\tau \int_{\tau}^{j} dt\; i(\tau) x(t) F(t-\tau) \\ &= \mathcal{R}_0 \int_{0}^{j} dt x(t) \int_{0^-}^{t} d\tau i(\tau) F(t-\tau) \\ &= \mathcal{R}_0 \int_{0^-}^{j} dt\; x(t) y(t) \\ &= \int_{0^+}^{j} dt\; i(t). \end{aligned}$$

Now we move on to calculate $C_2(j)$. We have

$$\begin{aligned}
C_2(j) &= \int_{0^-}^{j} d\tau \int_{\tau}^{j} d\rho \; w_j(\tau,\rho) \left(C(\tau,\rho)\right)^2 \\
&= \mathcal{R}_0^2 \int_{0^-}^{j} d\tau \int_{\tau}^{j} d\rho \; i(\tau) g_j(\rho,\tau) \left(\int_{\tau}^{\rho} dt x(t) \right)^2 \\
&= 2\mathcal{R}_0^2 \int_{0^-}^{j} ds \; x(s) \int_{0^-}^{s} dt \; x(t) \int_{0^-}^{t} d\tau \; i(\tau) F(s-t) F(t-\tau) \\
&= 2\mathcal{R}_0^2 \int_{0^-}^{j} ds \; x(s) \int_{0^-}^{s} dt \; x(t) y(t) F(s-t) \\
&= 2\mathcal{R}_0^2 \int_{0^-}^{j} ds \; x(s) \int_{0^-}^{s} dt \; x(t) y(t) F(s-t)
\end{aligned}$$

Following the same steps as in Section S.1.2, we obtain

$$\begin{aligned}
C_2(j) &= 2\mathcal{R}_0 \int_{0^-}^{j} ds \; x(s) \int_{0^+}^{s} dt \; i(t) F(s-t) \\
&= 2\mathcal{R}_0 \int_{0^-}^{j} ds \; x(s) \left(y(s) - y_0 F(s) \right) \\
&= 2\mathcal{R}_0 \int_{0^-}^{j} ds \; x(s) y(s) - 2\mathcal{R}_0 y_0 \int_{0^-}^{j} ds \; x(s) F(s) \\
&= 2 \int_{0^+}^{j} i(s) \; ds - 2\mathcal{R}_0 y_0 \int_{0}^{j} ds \; x(s) F(s) \\
&= 2C_1(j) - 2\mathcal{R}_0 y_0 \int_{0}^{j} ds \; x(s) F(s).
\end{aligned}$$

The second raw moment is *dependent* on the key parameter $\mathcal{R}_0$. However, as the initial cohort size y_0 tends to zero, the difference between $C_2(j)$ and twice the truncated first raw moment $2C_1(j)$ approaches zero. Apart from taking the limit of $C_2(j)$ as the initial infectious cohort size tends to zero, if we increase the truncation time j, the value of $C_1(j)$ increase, thereby the deviation introduced by y_0 diminishes. Hence, $\kappa(j)$, defined as

$$\kappa(j) = \frac{C_0(j) C_2(j)}{C_1^2(j)} - 1,$$

approaches 1.

Supplementary Figures

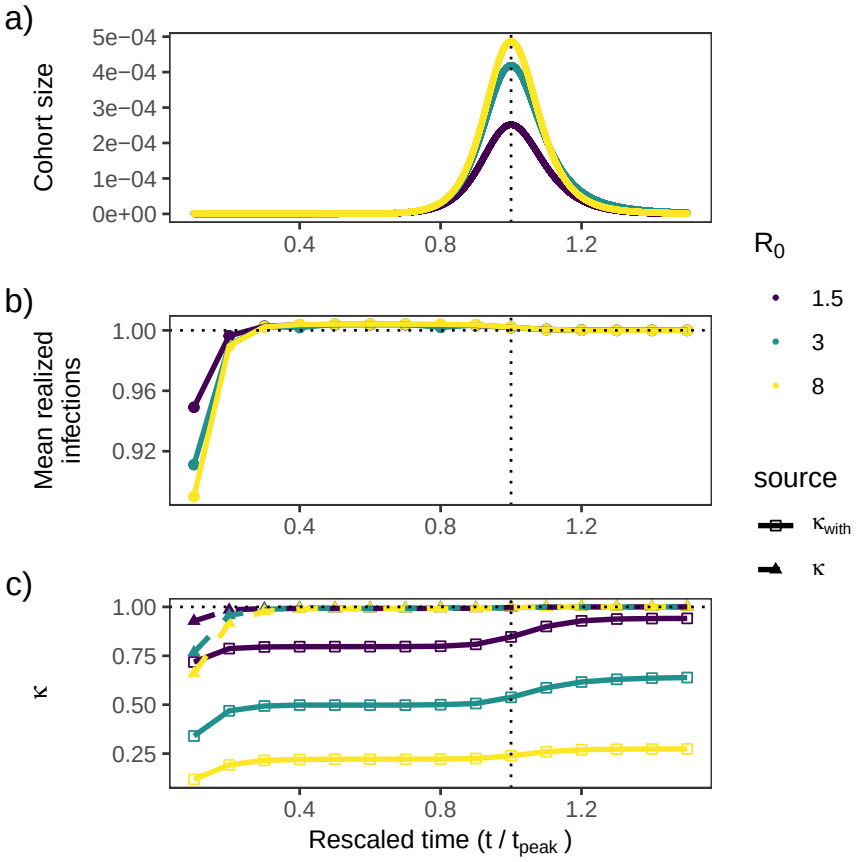


Figure S1: **Mean and heterogeneity of realized infections evolve over the course of an outbreak.** Panel a: The size of the cohort infected during $[t, t + \Delta t]$ is given by the area under the incidence curve over that interval, where $\Delta t = 10^{-4} t_{\text{peak}}$ and t_{peak} denotes the time of peak incidence. Panel b: At each time point, mean realized infections is computed using the cases had occurred by that time. Early in the outbreak, the initial infectious cohort constitutes a relatively large fraction of all infected population up to that time. Because we assume a "jump-start" outbreak, the infectors of this initial cohort are not included, and hence, the mean realized infections is noticeably below one. At early times ($t \lesssim 0.4 t_{\text{peak}}$), epidemics with larger $\mathcal{R}_0$ have accumulated fewer infections, making the effect of this omission more pronounced. Panel c: κ of realized infections increases over the course of the outbreak. For epidemics with smaller $\mathcal{R}_0$, a given fraction of t_{peak} corresponds to a longer elapsed time, allowing differences in infectious periods among cohort members to become apparent. As a result, the within component of κ (measured as the ratio of within-cohort variance to the squared mean) is larger than that for epidemics with larger $\mathcal{R}_0$. For panels b and c, the simulation was truncated at each plotted time. The y-axis value at each time point was computed by assigning to each cohort their realized number of infections up to that time. In all panels, time units measured in the mean infectious period are rescaled relative to the peak time, so that peak incidence occurs at $(t/t_{\text{peak}} = 1)$. The outbreaks were modeled using standardized compartmental models.

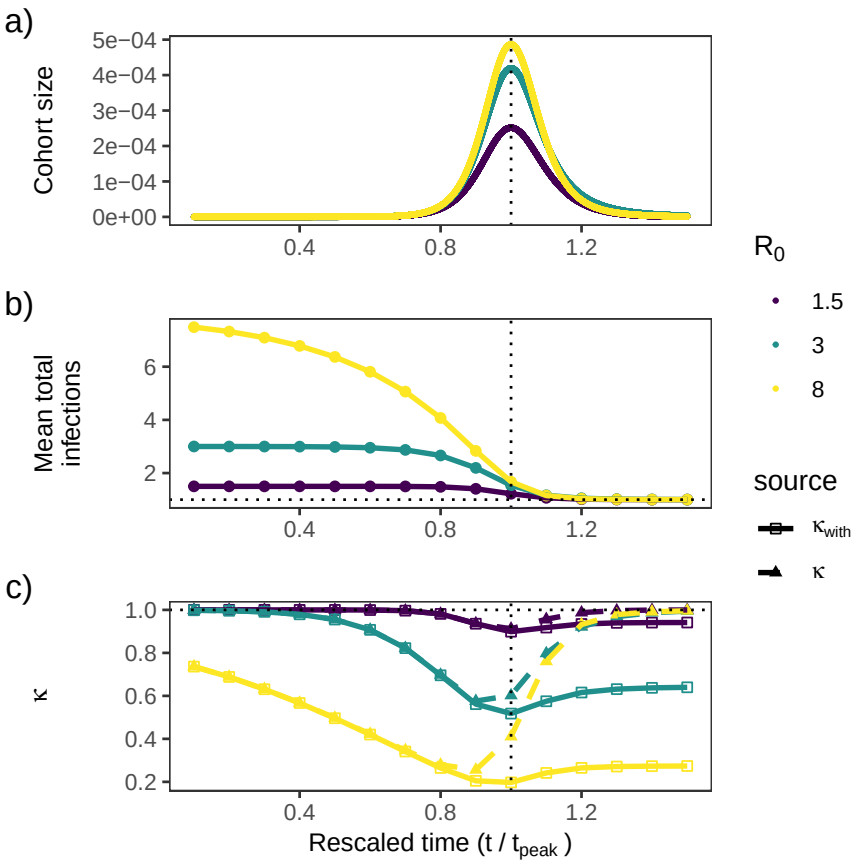


Figure S2: **Forecasts of the distribution of total infections evolve as an outbreak unfolds.** Panel a: Cohort size infected during $[t, t + \Delta t]$ is measured as the area under the incidence curve over this interval ($\Delta t = 10^{-4} t_{\text{peak}}$, where t_{peak} is the time of peak incidence). Panel b: At each time point, mean total infections is calculated across all cohorts infected by that time, including the initial infectious cohort of size 10^{-9}, while accounting for each cohort's subsequent contributions to disease transmission. Cohorts infected earlier have higher mean total infections, with values depending on key parameter $\mathcal{R}_0$. As incidence increases, later-infected cohorts, who have larger sizes, encounter susceptible pool depletion, decreasing mean total infections. After the peak, the susceptible population size stabilizes, and the mean approaches one. Panel c: The within-cohort component of the squared coefficient of variation κ_{with}, measured as the ratio of the within-cohort variance to the squared mean, is the main driver of heterogeneity among the early-infected cohorts, as they experience a similar size of susceptible population. As the outbreak unfolds, the decline in the susceptible population drives between-cohort variation, which is more pronounced in epidemics with large $\mathcal{R}_0$. κ reaches its minimum around the outbreak peak; large cohorts experience sharp depletion of susceptibles, and the impact of variation in recovery time fades, declining the variance, and hence κ. After the peak, κ rebounds to one. For panels b and c, the y-axis value at each plotted time point was quantified using only the cohorts that had been infected up to that time. The outbreaks were modeled using standardized compartmental models.

References

[1] Benjamin M Althouse, Edward A Wenger, Joel C Miller, Samuel V Scarpino, Antoine Allard, Laurent Hebert-Dufresne, and Hao Hu. Superspreading events in the transmission dynamics of SARS-CoV-2: Opportunities for interventions and control. *PLoS Biol*, 18:e3000897, 2020.

[2] David Champredon, Jonathan Dushoff, and David J. D. Earn. Equivalence of the Erlang-Distributed SEIR Epidemic Model and the Renewal Equation. *SIAM J. Appl. Math.*, 78:3258–3278, 2018.

[3] Paul Z Chen, Niklas Bobrovitz, Zahra Premji, Marion Koopmans, David N Fisman, and Frank X Gu. Heterogeneity in transmissibility and shedding SARS-CoV-2 via droplets and aerosols. *Elife*, 10, 2021.

[4] Akira Endo, Sam Abbott, Adam J Kucharski, and Sebastian Funk. Estimating the overdispersion in COVID-19 transmission using outbreak sizes outside China. *Wellcome Open Res*, 5:67, 2020.

[5] T Garske and C J Rhodes. The effect of superspreading on epidemic outbreak size distributions. *J Theor Biol*, 253:228–37, 2008.

[6] Ashish Goyal, Daniel B Reeves, and Joshua T Schiffer. Multi-scale modelling reveals that early super-spreader events are a likely contributor to novel variant predominance. *J R Soc Interface*, 19:20210811, 2022.

[7] Lea Hamner, Polly Dubbel, Ian Capron, Andy Ross, Amber Jordan, Jaxon Lee, Joanne Lynn, Amelia Ball, Simranjit Narwal, Sam Russell, Dale Patrick, and Howard Leibrand. High SARS-CoV-2 Attack Rate Following Exposure at a Choir Practice - Skagit County, Washington, March 2020. *MMWR Morb Mortal Wkly Rep*, 69:606–610, 2020.

[8] Herbert W. Hethcote. The Mathematics of Infectious Diseases. *SIAM Rev.*, 42:599–653, 2000.

[9] Alex James, Jonathan W Pitchford, and Michael J Plank. An event-based model of superspreading in epidemics. *Proc Biol Sci*, 274:741–7, 2007.

[10] William Ogilvy Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 115:700–721, 1927.

[11] Markus Kuba and Alois Panholzer. On moment sequences and mixed Poisson distributions. *Probab. Surveys*, 13:89–155, 2016.

[12] Jacob E Lemieux, Katherine J Siddle, Bennett M Shaw, Christine Loreth, Stephen F Schaffner, Adrianne Gladden-Young, Gordon Adams, Timelia Fink, Christopher H Tomkins-Tinch, Lydia A Krasilnikova, Katherine C

DeRuff, Melissa Rudy, Matthew R Bauer, Kim A Lagerborg, Erica Normandin, Sinead B Chapman, Steven K Reilly, Melis N Anahtar, Aaron E Lin, Amber Carter, Cameron Myhrvold, Molly E Kemball, Sushma Chaluvadi, Caroline Cusick, Katelyn Flowers, Anna Neumann, Felecia Cerrato, Maha Farhat, Damien Slater, Jason B Harris, John A Branda, David Hooper, Jessie M Gaeta, Travis P Baggett, James O'Connell, Andreas Gnirke, Tami D Lieberman, Anthony Philippakis, Meagan Burns, Catherine M Brown, Jeremy Luban, Edward T Ryan, Sarah E Turbett, Regina C LaRocque, William P Hanage, Glen R Gallagher, Lawrence C Madoff, Sandra Smole, Virginia M Pierce, Eric Rosenberg, Pardis C Sabeti, Daniel J Park, and Bronwyn L MacInnis. Phylogenetic analysis of SARS-CoV-2 in Boston highlights the impact of superspreading events. *Science*, 371, 2021.

[13] J O Lloyd-Smith, S J Schreiber, P E Kopp, and W M Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438:355–9, 2005.

[14] Michael T Meehan, Angus Hughes, Romain R Ragonnet, Adeshina I Adekunle, James M Trauer, Pavithra Jayasundara, Emma S McBryde, and Alec S Henderson. Replicating superspreader dynamics with compartmental models. *Sci Rep*, 13:15319, 2023.

[15] Bjarke Frost Nielsen, Kim Sneppen, and Lone Simonsen. The counterintuitive implications of superspreading diseases. *Nat Commun*, 14:6954, 2023.

[16] Lorenzo Pellis, Paul J Birrell, Joshua Blake, Christopher E Overton, Francesca Scarabel, Helena B Stage, Ellen Brooks-Pollock, Leon Danon, Ian Hall, Thomas A House, Matt J Keeling, Jonathan M Read, and Daniela De Angelis. Estimation of reproduction numbers in real time: Conceptual and statistical challenges. *J R Stat Soc Ser A Stat Soc*, 185:S112–S130, 2022.

[17] D Schenzle. An age-structured model of pre- and post-vaccination measles transmission. *IMA J Math Appl Med Biol*, 1:169–91, 1984.

[18] Kim Sneppen, Bjarke Frost Nielsen, Robert J Taylor, and Lone Simonsen. Overdispersion in COVID-19 increases the effectiveness of limiting nonrepetitive contacts for transmission control. *Proc Natl Acad Sci U S A*, 118, 2021.

[19] M E Woolhouse, C Dye, J F Etard, T Smith, J D Charlwood, G P Garnett, P Hagan, J L Hii, P D Ndhlovu, R J Quinnell, C H Watts, S K Chandiwana, and R M Anderson. Heterogeneities in the transmission of infectious agents: implications for the design of control programs. *Proc Natl Acad Sci U S A*, 94:338–42, 1997.